Syllabus

Course Contents	
Unit I	Introduction to Networking: Define Network, Computer Network, Goals and Applications of Networks, Organization of the Internet, ISP, Network Structure and Architecture (Layering Principles, Services, Protocols and Standards), The OSI reference model, TCP/IP Protocol Suite, Network Devices and Components. [5 Hrs]
Unit II	Network Access & Tools: Ethernet LANs and Switches, VLANs and Trunking, Wireless LANs, Network Access Control, Basic Device Configuration. Network Tools: OPNET, IPERF, JPERF, Wireshark, DummyNet Emulator. [5 Hrs]
	Communication Networking Techniques:

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Unit III	Circuit Switching, Packet Switching, Local Area Networks. Protocols, Layered Approach, System Network Architecture. [5 Hrs]

Text Books

1.	Data Communications and Networking, by Behrouz A. Forouzan, McGraw Hill , 2006
2.	"Computer Networks",Andrew Tanenbaum , Prentice Hall.
3	Network Fundamentals: CCNA Exploration Companion Guide (CiscoNetworking Academy).
4	Computer Networks ,A.S. Tanenbaum, , PHI , 2002.

Reference Books

1.	Computer Networks: A Systems Approach, by Larry Peterson and Bruce Davie. Covers background networking material that students should already be familiar with.
2.	Computer Networking: A Top-Down Approach Featuring the Internet, by James F. Kurose and Keith W. Ross. Covers similar material to Peterson and Davie.
3.	TCP/IP Illustrated, Volume 1: The Protocols by W. Richard Stevens.
4.	Unix Network Programming: Networking APIs: Sockets and XTI (Volume 1) by W. Richard Stevens.

Useful Links

1	https://www.google.co.in/books/edition/Network_Management_Fundamentals/NvEVQmjScjYC?hl=en&gbpv=1&bsq=networking+tools+%26+techniques+syllabus&dq=networking+tools+%26+techniques+syllabus&printsec=frontcover
2	https://www.google.co.in/books/edition/Computer_Networks/BvaFreun1W8C?hl=en&gbpv=1&dq=computer+networking&printsec=frontcover

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